

CNN-15 MODEL REPORT

CLASSIFICATION ACCURACY ASSESSMENT REPORT

Model : CNN-15
Image File : /content/drive/My Drive/Internship_project/Singrauli_LULC_CNN15_2024.tif
User Name : Danesh
Date : Wed Feb 11 06:23:16 2026

ERROR MATRIX

		Reference Data					

Classified Data							

	Water	Agriculture	Settlement	Mining	Barren	Forest	
Water	25	1	0	0	0	0	
Agriculture	2	13	5	0	6	2	
Settlement	0	5	7	0	1	0	
Mining	0	0	2	13	0	0	
Barren	1	5	1	1	3	0	
Forest	0	0	2	0	1	24	
		Water	Agriculture	Settlement	Mining	Barren	Forest
Row Total	26	28	13	15	11	27	
Column Total	28	24	17	14	11	26	

----- End of Error Matrix -----

ACCURACY TOTALS

Class Name	Reference Totals	Classified Totals	Number Correct	Producers Accuracy	Users Accuracy
Water	28	26	25	89.29%	96.15%
Agriculture	24	28	13	54.17%	46.43%
Settlement	17	13	7	41.18%	53.85%
Mining	14	15	13	92.86%	86.67%
Barren	11	11	3	27.27%	27.27%
Forest	26	27	24	92.31%	88.89%

Overall Classification Accuracy = 70.83%

KAPPA (K[^]) STATISTICS

Overall Kappa Statistics = 0.6424

Class Name Kappa
Water 0.8632
Agriculture 0.4022
Settlement 0.3403
Mining 0.9184
Barren 0.1993
Forest 0.9007

----- End of Kappa Statistics -----

CNN-9 MODEL REPORT

CLASSIFICATION ACCURACY ASSESSMENT REPORT

Model : CNN-9
Image File : /content/drive/My Drive/Internship_project/Singrauli_LULC_CNN9_2024.tif
User Name : Danesh
Date : Wed Feb 11 06:23:16 2026

ERROR MATRIX

		Reference Data					

Classified Data							

	Water	Agriculture	Settlement	Mining	Barren	Forest	
Water	24	1	0	1	0	0	
Agriculture	2	14	3	0	7	2	
Settlement	0	6	6	0	1	0	
Mining	0	0	2	12	1	0	
Barren	1	5	1	1	3	0	
Forest	0	0	2	0	2	23	
	Water	Agriculture	Settlement	Mining	Barren	Forest	
Row Total	26	28	13	15	11	27	
Column Total	27	26	14	14	14	25	

----- End of Error Matrix -----

ACCURACY TOTALS

Class Name	Reference Totals	Classified Totals	Number Correct	Producers Accuracy	Users Accuracy
Water	27	26	24	88.89%	92.31%
Agriculture	26	28	14	53.85%	50.00%
Settlement	14	13	6	42.86%	46.15%
Mining	14	15	12	85.71%	80.00%
Barren	14	11	3	21.43%	27.27%
Forest	25	27	23	92.00%	85.19%

Overall Classification Accuracy = 68.33%

KAPPA (K[^]) STATISTICS

Overall Kappa Statistics = 0.6119

Class Name Kappa
Water 0.8582
Agriculture 0.3980
Settlement 0.3591
Mining 0.8367
Barren 0.1350
Forest 0.8968

----- End of Kappa Statistics -----

Random Forest MODEL REPORT

CLASSIFICATION ACCURACY ASSESSMENT REPORT

Model : Random Forest
Image File : /content/drive/My Drive/Internship_project/Singrauli_LULC_RF_2024.tif
User Name : Danesh
Date : Wed Feb 11 06:23:16 2026

ERROR MATRIX

		Reference Data					

Classified Data							

	Water	Agriculture	Settlement	Mining	Barren	Forest	
Water	20	0	0	6	0	0	
Agriculture	0	16	5	0	7	0	
Settlement	0	2	11	0	0	0	
Mining	0	0	1	14	0	0	
Barren	0	0	0	0	11	0	
Forest	0	2	3	0	2	20	
	Water	Agriculture	Settlement	Mining	Barren	Forest	
Row Total	26	28	13	15	11	27	
Column Total	20	20	20	20	20	20	

----- End of Error Matrix -----

ACCURACY TOTALS

Class Name	Reference Totals	Classified Totals	Number Correct	Producers Accuracy	Users Accuracy
Water	20	26	20	100.00%	76.92%
Agriculture	20	28	16	80.00%	57.14%
Settlement	20	13	11	55.00%	84.62%
Mining	20	15	14	70.00%	93.33%
Barren	20	11	11	55.00%	100.00%
Forest	20	27	20	100.00%	74.07%

Overall Classification Accuracy = 76.67%

KAPPA (K^) STATISTICS

Overall Kappa Statistics = 0.7200

Class Name	Kappa
Water	1.0000
Agriculture	0.7391
Settlement	0.4953
Mining	0.6571
Barren	0.5046
Forest	1.0000

----- End of Kappa Statistics -----